MIL-B-27036A(USAF)
9 February 1962
Superseding
MIL-B-27036(USAF)
20 November 1961

MILITARY SPECIFICATION

BOMB, LOADING PRACTICE BDU-9/E

1. SCOPE. This specification covers the requirements for one type of loading practice bomb designated as the BDU-9/E, hereafter referred to as the practice bomb.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-A-468	Aluminum - Pigment: Powder and Paste (For) Paint
TT-T-266	Thinner: Dope and Lacquer (Cellulose-Nitrate)
PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock - Corner
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-V-3083	Vibrator, Concrete, Pneumatic
MIL-P-6889	Primer; Zinc-Chromate, For Aircraft Use
MIL-L-7178	Lacquer; Cellulose-Nitrate, Gloss, For Aircraft Use
MIL-T-9107	Test Reports, Preparation Of
MIL-P-11414	Primer; Lacquer, Rust-Inhibiting
MIL-B-27000	Bombs, Practice, General

FSC 1115

AF Hooven-Dayton Co., 21 March 1962 1000

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Air Force

60J45674	Bomb, Loading Practice BDU-9/E, Assembly Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION

3.1.1 PREPRODUCTION TESTING. This specification makes provisions for preproduction testing.

3.2 DESIGN AND CONSTRUCTION. The practice bomb shall be designed and constructed in conformance to drawing number 60J45674 and subsidiary drawings.

3.3 CONCRETE FILL

3.3.1 GENERAL. The practice bomb shall contain concrete fill, provided to accomplish the function of ballast, and to allow accurate weight simulation of the parent weapon. The concrete fill process shall be the responsibility of the contractor on all bombs.

3.3.2 DENSITY OF CONCRETE. The concrete shall have a specific density of 143 pounds per cubic foot. The allowable tolerance for the density of the concrete shall be plus 7 and minus 18 pounds per cubic foot.

3.3.3 MIXING OF CONCRETE. The procedure for mixing the concrete shall be as recommended in MIL-B-27000.

3.3.4 METHOD OF FILLING

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3.3.4.1 **BALLAST SECTION.** The ballast section shall be filled with concrete in the following manner:

a. Position the bomb body with forward end up.

b. Fill with concrete to 1/4 the distance to the fill line.

c. Insert pneumatic vibrator conforming to MIL-V-3083 and vibrate for a period of from 30 to 60 seconds.

d. Add concrete to 1/2 the distance to the fill mark and repeat step c.

e. Add concrete to 3/4 the distance to the fill mark and repeat step c.

f. Add concrete to the fill line and repeat step c.

3.3.4.2 **SIMULATED PARACHUTE CONTAINER.** The simulated parachute container shall be filled with concrete in the following manner:

a. Position the simulated parachute container with the forward end down.

b. Fill with concrete to the fill mark.

c. Repeat step c. of 3.3.4.1.

3.3.4.3 CURING TIME. Curing time for the concrete fill shall be as specified in MIL-B-27000.

3.4 FINISH. Finish of the bomb shall be in accordance with MIL-B-27000. Surfaces of parts of aluminum or ferrous metal requiring lacquer finish shall receive one coat of primer, MIL-P-11414. Surfaces of parts of aluminum or ferrous metal not requiring lacquer finish shall receive one coat of primer, MIL-P-6889. All finish parts receiving a finish coat of lacquer shall be allowed to air dry for a minimum of one hour before handling.

3.4.1 EXTERNAL SURFACES. All external surfaces of the practice bomb shall receive one coat of primer as specified in paragraph 3.4. After the surface is dry, one coat of lacquer shall be applied. The lacquer to be used shall be a mixture of clear lacquer as specified in MIL-L-7178, and aluminum pigment as specified in TT-A-468, Type 2, Grade A. The lacquer shall be thinned by one volume of lacquer thinner, TT-T-266, to two volumes of lacquer.

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3.4.2 SIMULATED WARHEAD. The simulated warhead shall receive the finish coat treatment (external surfaces) in a manner identical to that described in paragraph 3.4.1 for finishing external surfaces.

3.4.3 SIMULATED PARACHUTE CONTAINER. The simulated parachute container shall receive one coat of primer as specified in paragraph 3.4. After the surface is dry, one coat of lacquer shall be applied. The lacquer to be used shall be as specified in MIL-L-7178, using color number 1405 (olive drab) of FED-STD-595. The lacquer shall be thinned by one volume of lacquer thinner, TT-T-266, to two volumes of lacquer.

3.5 PERFORMANCE

3.5.1 FUNCTIONAL. The functional performance of the practice bomb shall be interpreted as proper performance of the Arm-Safe circuit, the Lay-Down Timer, and the pressure sensing circuit.

3.5.2 STATIC LOADS. The practice bomb shall be capable of withstanding the static loads resulting from suspension, handling, loading, and transportation.

3.5.3 SHOCK LOADS. The practice bomb shall be capable of meeting all functional performance requirements of this specification after being subjected to the shock loads specified in MIL-B-27000.

3.6 IDENTIFICATION OF PRODUCT. All equipment, assemblies, and parts of the practice bomb shall be marked for identification as specified in MIL-STD-130.

3.7 The following items of equipment will be loaned to the contractor, upon his request, as Government-loaned equipment for the purpose of testing the practice bomb:

ITEM NO.	PART NUMBER	DESCRIPTION OF ITEM	QUANTITY
a.	NOCM1190-5460553	Adapter Hose to Valve T-298	1
b.	NOCM6685-575-4692	Guage, Pressure, Dial Indicating T-283	1

3.8 WORKMANSHIP. Workmanship shall be in accordance with MIL-B-27000.

4. QUALITY ASSURANCE PROVISIONS

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4.1 CLASSIFICATION OF TESTS. Inspection and testing of the practice bomb shall consist of the following:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall be representative of the production practice bombs. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 PREPRODUCTION TEST REPORT. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described under "4.5 TEST METHODS."

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests.

a. Examination of product.

b. Conformance to drawings.

c. Continuity Test.

d. Pressure Test.

4.4 TEST CONDITIONS. Test conditions shall be as described for each specific test.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each practice bomb shall be examined for conformance to the requirements of this specification, with the exception of the performance requirements. Non-conformance with requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each practice bomb shall be inspected for conformance to the drawings, specifications, and standards specified therein. Failure to conform to these criteria shall be cause for rejection.

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4.5.3 CONTINUITY TEST. Using any suitable instrumentation, the electrical circuits shown in Figure 1 shall be subjected to a continuity test, with the switch in each position. Failure to meet the requirement of this test shall be cause for rejection.

4.5.4 PRESSURE TEST. Pressurize the pressure tube to 15 pounds guage. After a period of 96 hours, inspect for pressure by means of a T-289 Pressure Tester attached to pressure valve of the simulated warhead. An indicated pressure of less that 12 pounds shall be cause for rejection.

4.5.5 STATIC LOAD TESTS

4.5.5.1 TEST SET UP. By means of two straps three inches in width, the practice bomb, after concrete fill is completed, shall be suspended from a test frame. The straps and test frame shall be capable of withstanding the loads imposed upon the practice bombs. The straps shall be positioned so that the rear edge of one will be at station 101.95 and the forward edge of the other will be at station 34.00.

NOTE: Station 2.00 is located at the forward edge of the nose. Static Loads shall be applied through flexible bands of sufficient width to prevent local compression failure.

4.5.5.2 TEST PROCEDURE. Test procedure shall be as specified in Procedure I of the Suspension Test described in MIL-B-27000. The design limit loads are defined below:

a. In the vertical plane:

Station 52.0	down	28,500 pounds
92.0	down	28,500 pounds

b. In the horizontal plane:

Station 72.25	left	15,200 pounds

c. In the fore and aft plane through straps parallel to the center line of the practice bomb, applied to a plate which shall be bolted to the aft end of the practice bomb after removal of the aft cover and the simulated parachute container:

Total forward load	57,000 pounds

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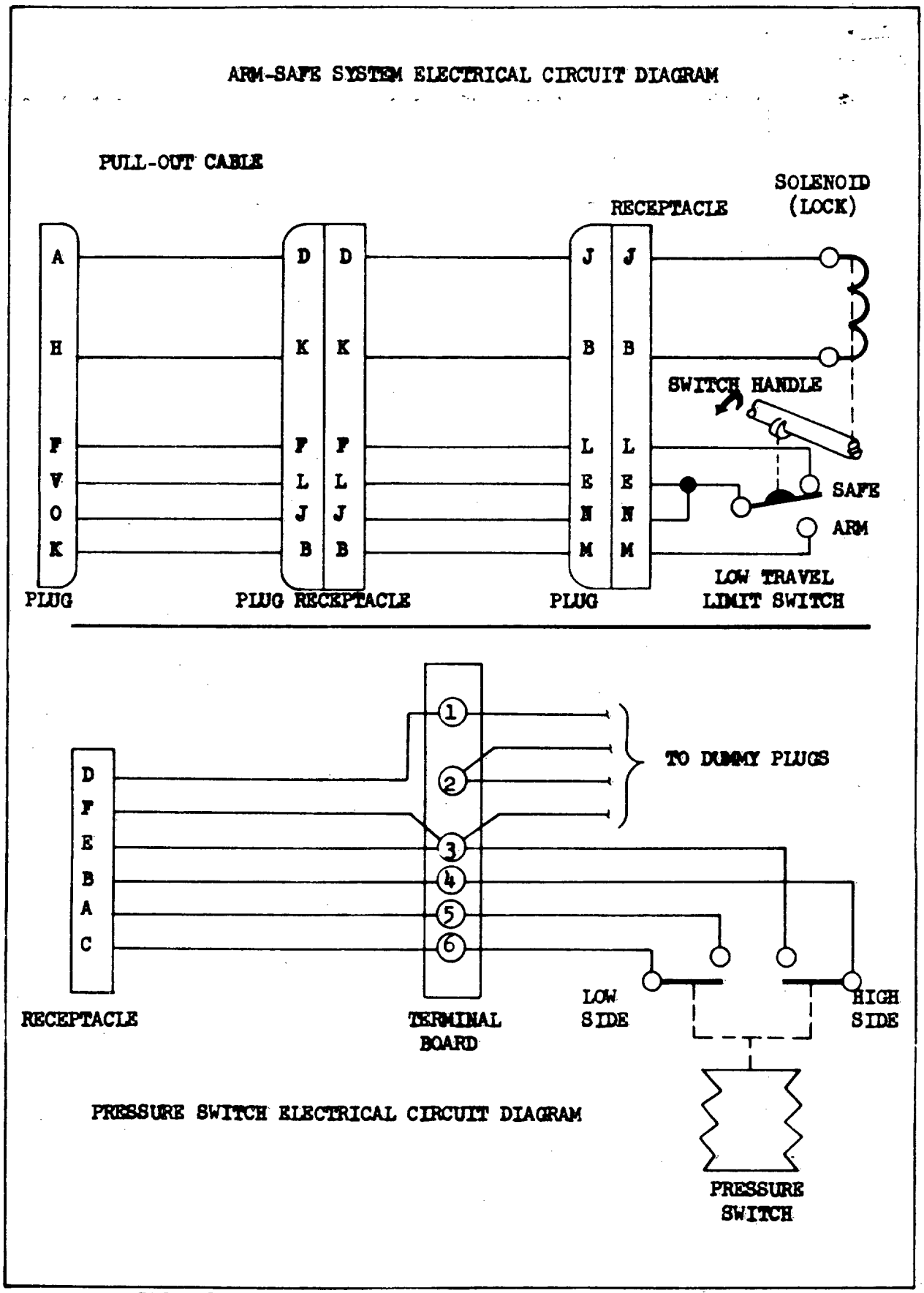


FIGURE 1. PRACTICE BOMB ELECTRICAL CIRCUIT DIAGRAM

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After meeting the requirements of the test specified in MIL-B-27000, the practice bomb shall be subjected to the continuity and pressure tests. Failure to pass either of these tests shall be considered as failure to pass the Static Load Test and cause for rejection.

4.5.6 SHOCK TEST

4.5.6.1 TEST SET UP. Set up conditions for the shock test are to be as follows:

a. The practice bomb shall be filled with concrete as specified, and shall contain the simulated parachute container.

b. The test fixture utilized shall be in accordance with MIL-B-27000 with the exception that the 30 inch suspension system simulation and sway brace criteria does not apply. The simulated suspension shall have two straps located at the stations specified in the static load test set up.

4.5.6.2 TEST PROCEDURES. Test procedures for the shock test shall be as specified in MIL-B-27000. Failure or deformation of any portion of the practice bomb shall be cause for rejection. Subsequent to meeting the above requirements, the practice bomb shall be subjected to the continuity and pressure tests. Failure to pass either of these tests shall be considered as failure to pass the shock test and cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION. For all levels of shipment, all machined surfaces and surfaces not having a protective final finish shall be preserved in accordance with Method I, MIL-P-116.

5.2 PACKAGING, ALL LEVELS. The fins, pullout cables and arming wires shall be removed from the practice bomb and individually wrapped with barrier material conforming to MIL-B-131. The barrier material shall be sealed with tape conforming to PPP-T-60. Attaching hardware shall be sealed in a waterproof bag. The fins, pullout cables, arming wires and attaching hardware shall be packed in a box conforming to PPP-B-601 or PPP-B-621. Physical protection shall be in accordance with JAN-P-100.

5.3 PACKING, ALL LEVELS. The practice bomb body, containing the simulated parachute container, shall be covered with barrier material conforming to MIL-B-131, sealed with tape conforming to PPP-T-60 and securely tied down to a Trailer, Dummy Bomb MHU-49/E.

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5.4 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. Shipment marking nomenclature shall be:

BOMB, LOADING PRACTICE BDU-9/E

5.4.1 ADDITIONAL MARKING. In addition to marking in accordance with MIL-STD-129, the following notice shall appear prominently on the exterior of barrier material covering the nose and rear plate of the practice bomb: FRAGILE: DO NOT PUSH THIS AREA.

6. NOTES

6.1 The Bomb, Loading Practice BDU-9/E is intended for use in training ground crews in the handling and loading of the parent weapon. The practice bomb is non-expendable and not intended to be dropped from the aircraft.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government-loaned property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procuring activity. (Section 4.2.1)

d. Levels for preservation and packing.

6.2.1 PREPRODUCTION TEST SAMPLES. It is expected the contract or purchase order will specify that a minimum of two units representative of the production Bomb, Loading Practice BDU-9/E will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contractor purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production item should be delivered prior to approval of the preproduction samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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